CORPORATE IT SUPPORT AND WORKSTATION EFFICIENCY

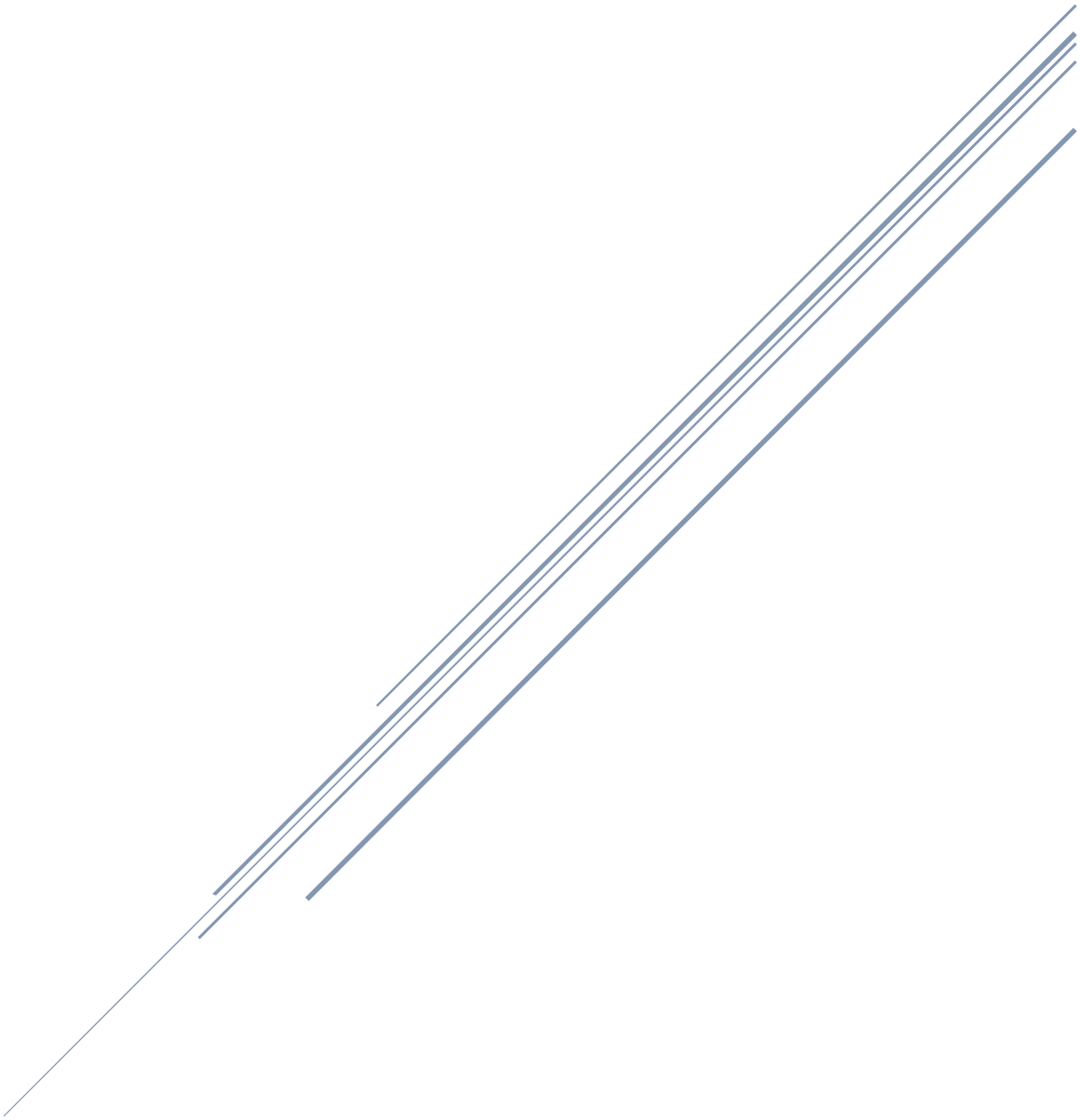


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1. Operating System Maintenance

System maintenance is a crucial aspect of ensuring that your operating system (OS) runs smoothly, efficiently, and securely. In this document, we will explore the definition and importance of system maintenance, the types of tasks involved, and the benefits of regular maintenance. System maintenance refers to the process of performing routine tasks to keep your operating system and its components in good working condition.

It helps with:

- Preventing system crashes and data loss
- Improving system performance and speed
- Enhancing security by patching vulnerabilities
- Ensuring compatibility with new software and hardware

1. System Optimization

Optimizing your OS improves speed and efficiency. Key steps include:

- Removing unnecessary files and programs
- Cleaning temporary files and browser caches
- Defragmenting hard drives (if applicable)
- Cleaning the system registry
- Disabling unnecessary startup programs
- Adjusting power settings for performance or battery life
- Using disk cleanup or optimization tools to automate maintenance

1.1. Install updates weekly:

Keeping your operating system and applications up to date is essential. Updates often include security patches, bug fixes, and performance improvements. Users should enable automatic updates where possible, or regularly check for updates manually. For Windows, this can be done via Settings > Update & Security > Windows Update, and for macOS, through System Preferences > Software Update, ensuring automatic updates are enabled.

1.2. Run antivirus scans: Follow organization policy and document changes.

Protecting your OS from viruses and malware is critical. Install reputable antivirus or anti-malware software, keep it updated, and perform regular scans. This helps prevent data loss, privacy breaches, and system malfunctions

1.3. Remove unused software:

Review installed programs monthly and uninstall anything you no longer use. This reduces background processes, frees storage, and improves startup times.

On **Windows**, use *Settings* → *Apps* → *Installed apps*; on **macOS**, drag apps to the Trash or use Launchpad; on **Linux**, use your package manager (apt remove, dnf remove, etc.).

Avoid removing system-critical apps; if unsure, search the app name before deleting.

1.4. Monitor storage space:

Check your disk usage regularly to prevent slowdowns and unexpected "low storage" warnings.

Windows: Settings → System → Storage shows usage by category.

macOS: Apple Menu → About This Mac → Storage.

Linux: Use df -h or a graphical tool like Disk Usage Analyzer.

Set alerts or reminders if your system supports them—performance drops sharply when storage falls below 10–15% free space.

1.5. Check startup programs:

Disabling unnecessary startup applications reduces boot time and frees system resources. On Windows, this can be done via

Task Manager > Startup tab, and on macOS through System Preferences > Users & Groups > Login Items

By following these steps consistently, users can maintain their operating system effectively, ensuring it remains secure, fast, and reliable over time.

1.6. Clean temporary files: Follow organization policy and document changes

Temporary files accumulate from updates, browsers, installers, and apps; clearing them improves speed and frees space.

Windows: Use *Disk Cleanup* or *Storage Sense* to automatically remove temp files, recycle bin contents, and old update files.

macOS: Use *Manage Storage* → *Reduce Clutter* or clear caches via Finder (~/Library/Caches).

Linux: Use sudo apt autoremove, sudo apt clean, or tools like *BleachBit* (used carefully).

Clear browser caches periodically if you browse heavily.

1.7. Verify backups:

Regularly backing up important files protects against data loss from hardware failure, malware, or accidental deletion. Users can use external drives, cloud storage, or network devices. It is important to test backups periodically to ensure they are recoverable. For comprehensive protection, consider full system backups, especially for virtual machines or critical systems

1.8. Hardware Maintenance

Physical maintenance ensures longevity and proper functioning of your system. Steps include:

- Cleaning dust from internal components and peripherals
- Ensuring proper airflow to prevent overheating
- Checking and maintaining cooling systems
- Inspecting and cleaning keyboards, monitors, and other devices

1.9 Monitoring and Troubleshooting

Regularly monitor system performance and address issues promptly. Close resource-heavy programs if the system slows down, review error reports, and troubleshoot problems as they arise. Adaptive maintenance, such as upgrading hardware or software to meet changing requirements, also helps maintain optimal performance.

1.10 Running Disk Cleanups and Defragmentation

Disk cleanups and defragmentation are essential maintenance tasks that help to improve system performance. Disk cleanups remove unnecessary files, while defragmentation rearranges data on your disk to improve access

Disk cleanup in Windows

To delete temporary files:

1. In the search box on the taskbar, type **disk cleanup**, and select **Disk Cleanup** from the list of results.
2. Select the drive you want to clean up, and then select **OK**.
3. Under **Files to delete**, select the file types to get rid of. To get a description of the file type, select it.
4. Select **OK**.

If you need to free up more space, you can also delete system files:

5. In Disk Cleanup, select **Clean up system files**.
6. Select the file types to get rid of. To get a description of the file type, select it.
7. Select **OK**.

2. Installing key software:

2.1 Microsoft 365

You install Microsoft 365 by signing in at m365.cloud.microsoft/apps, selecting Install apps, downloading the installer, and running it on your device.

What you need before installing

- A Microsoft account or work/school account linked to your Microsoft 365 subscription
- If it's your first time, you may need to redeem a product key at *microsoft365.com/setup*
- Make sure your device meets system requirements (Windows or macOS)

1 Source

💻 Install Microsoft 365 on Windows or Mac

All official Microsoft instructions point to the same process

Microsoft+1

1. Go to the Microsoft 365 apps portal
   Visit https://m365.cloud.microsoft/apps
2. If needed, you can also use office.com or account.microsoft.com/services
3. Sign in
   - Use the Microsoft account associated with your subscription (Personal/Family)
   - Or use your work/school account if it's a business subscription
4. Download the installer
   - Select Install apps → Install
   - For work/school accounts, choose Microsoft 365 apps
5. Run the installer
   Depending on your browser
   - Edge/IE: Run
   - Chrome: Setup
   - Firefox: Save File, then open it

When prompted with *“Do you want to allow this app to make changes to your device?”*, select Yes

6. Wait for installation to finish
   You’ll see *“You're all set! Office is installed now”* when complete

✅ Activate Microsoft 365

1. Open Word, Excel, or any app from the Start menu.
2. Accept the license agreement.
3. The app activates automatically using your signed-in account

If activation fails, the Activation Wizard will guide you through fixing it

🔑 If you have a product key

If your Microsoft 365 came with a key, redeem it first at microsoft365.com/setup

Microsoft

This links the license to your Microsoft account so you can install anytime.

👨‍💼 If you use Microsoft 365 for business

- You must have a license assigned by your organization before you can install
- After that, installation works the same way via m365.cloud.microsoft/apps.

❗ If you don’t see an Install button

Microsoft notes this usually means an account issue—use the Need help? section in the portal for fixes

2.2 Microsoft Teams

If you prefer not to install anything, you can also use Teams directly in your browser at https://teams.microsoft.com

📥 Download Microsoft Teams (Official Source)

The official Microsoft download page provides installers for Windows, macOS, iOS, Android, and Linux
https://www.microsoft.com/microsoft-teams/download-app

Microsoft notes that the classic Teams app is being replaced by the new Teams app, which is faster and more secure

🖥️ Install on Windows

- Visit teams.microsoft.com/downloads.
- Click Download for desktop.
- Open the downloaded installer (.exe).
- Follow the on-screen setup instructions.
- Sign in with your Microsoft account.

You can also install via the Microsoft Store:

- Open Microsoft Store.
- Search Microsoft Teams.
- Select the app and click Install

🍏 Install on macOS

- Go to teams.microsoft.com/downloads.
- Select Download for macOS.
- Open the .pkg file.
- Follow installation prompts.
- Launch Teams and sign in.

📱 Install on iPhone or iPad

- Open the App Store.
- Search Microsoft Teams.
- Tap Get.
- Open the app and sign in.

🤖 Install on Android

- Open the Google Play Store.
- Search Microsoft Teams.
- Tap Install.
- Launch the app and sign in.

🐧 Install on Linux (Ubuntu, Debian, Fedora)

- Visit teams.microsoft.com/downloads.
- Download the .deb (Debian/Ubuntu) or .rpm (Fedora) package.
- Open it with your system's package manager.
- Complete installation and sign in.

💻 Install on Chromebook

- Open the Google Play Store.
- Search Microsoft Teams.
- Tap Install.
- Sign in.
  Or use the web version at teams.microsoft.com.

🌐 Use Teams Without Installing

Microsoft confirms you do not need the app to use Teams

- Go to https://teams.microsoft.com
- Sign in and use Teams in your browser.

🔑 Account Requirements

Microsoft Support notes:

- You need a Microsoft account for Teams Free
- Work/school accounts require the correct Microsoft 365 license

2.3 Google Chrome

You can install Google Chrome by downloading the installer from Google's official site on a computer or by installing it from the Play Store or App Store on mobile devices.

Below is a clear, device-by-device guide using official Google instructions.

💻 Install Chrome on Windows (Windows 10 and up)

- Download the Chrome installation file from Google's official site
- When prompted with "Do you want to run this file?", select Run
- If Windows asks "Allow this app to make changes?", select Yes
- Chrome will install automatically, and a Chrome window opens when done

Tip: Chrome is free and updates automatically every four weeks

🖥️ Install Chrome on macOS (macOS 13 Ventura and up)

- Download the googlechrome.dmg file from Google's site
- Open the downloaded file and drag Chrome into the Applications folder
- You may need to enter your Mac's admin password
- Eject the installer when finished to clean up

📱 Install Chrome on Android (Android 10 and up)

- Open the Google Play Store on your device
- Search for Chrome.
- Tap Install, then Open to start browsing

Note: Many Android devices already have Chrome preinstalled.

📱 Install Chrome on iPhone or iPad

- Open the App Store.
- Search for Google Chrome.
- Tap Get, then confirm with Face ID, Touch ID, or your Apple ID password
- Once installed, tap the Chrome icon to open it.

🖥️ Install Chrome on Linux (Debian/Ubuntu/Fedora)

- Download the .deb or .rpm package from Google's site
- Double-click the file to install via your system's package manager.

✅ Before You Install

Google recommends checking system requirements first

- Ensure your operating system is supported.
- Make sure you have enough disk space (Chrome typically needs a few hundred MB).

3. Hardware Troubleshooting:

1. Device won't power on

- Check power cable seating
- Try another outlet.
- Inspect PSU switch.
- Test with another power cable.

🔋 2. Power supply suspected faulty

- Use a PSU tester.
- Check voltage output
- Replace PSU if unstable.

1 Source

🔌 3. Laptop not charging

- Test with another charger.
- Inspect charging port for debris.
- Remove battery (if removable) and reseat.

🖥️ 4. No display on monitor

- Verify monitor cable connection
- Test with another monitor.
- Reseat GPU.
- Reseat RAM (display issues often RAM-related

🖥️ 5. Monitor says "No Signal"

- Ensure GPU is fully seated.
- Try another video port.
- Reset monitor input source.

🔊 6. No sound from speakers

- Check audio output device
- Inspect cable/port.
- Test left/right balance

🎤 7. Microphone not detected

- Check permissions in OS/browser
- Ensure mic is not muted.
- Reinstall audio drivers.

📹 8. Webcam not working

- Check browser permissions
- Close apps using the camera.
- Update camera drivers.

🖱️ 9. Mouse double-clicking

- Test on another PC
- Adjust double-click speed.
- Clean or replace switches.

⌨️ 10. Keyboard keys not responding

- Try another USB port
- Check layout settings.
- Reinstall keyboard drivers.

💾 11. Hard drive making noises

- Backup immediately.
- Run CHKDSK or SMART test
- Replace drive if errors persist.

💽 12. SSD not detected

- Check SATA/NVMe seating.
- Enable drive in BIOS
- Update motherboard firmware.

🧠 13. RAM errors or crashes

- Run MemTest86
- Reseat RAM.
- Test sticks individually.

🔥 14. Overheating

- Clean dust from fans
- Reapply thermal paste.
- Improve airflow.

🌀 15. Fan not spinning

- Check fan header connection
- Test with another header.
- Replace fan.

🧊 16. CPU overheating

- Ensure cooler is mounted correctly.
- Check pump (AIO).
- Replace thermal paste.

🎮 17. GPU overheating

- Clean heatsink.
- Increase fan curve.
- Replace thermal pads.

🧩 18. Hardware conflict

- Check Device Manager for warnings
- Update drivers.
- Remove conflicting devices.

🦠 19. Suspected malware causing hardware-like issues

- Run full malware scan
- Remove suspicious startup apps.

🖨️ 20. Printer not printing

- Clear print queue
- Reinstall printer.
- Check ink/toner levels.

📶 21. Wi-Fi keeps disconnecting

- Reduce interference
- Update Wi-Fi drivers.
- Restart router.

🌐 22. High ping or jitter

- Switch to Ethernet
- Close background apps.
- Reset network adapter.

🧱 23. Dead or stuck pixels

- Identify pixel type
- Use pixel-healing software.
- Replace monitor if dead pixel persists.

🔄 24. System freezes

- Check for overheating

- Run RAM test.
- Update drivers.

🧹 25. Slow computer

- Remove startup bloat
- Free disk space.
- Check for overheating.

🧲 26. USB ports not working

- Test another device.
- Check BIOS USB settings
- Reinstall USB controller drivers.

🔧 27. Loose internal cable

- Open case
- Reseat SATA, power, and front-panel connectors.

🧯 28. Burning smell or scorch marks

- Power off immediately.
- Inspect motherboard for burnt components
- Replace damaged parts.

🧪 29. Run built-in diagnostics

- Use OEM tools like Dell PSA
- Record error codes.

📝 30. Document the problem and fix

- Record symptoms and steps taken
- Update internal documentation.
- Share findings with team.